Orion

An AI Business Operating System. Five departments generate a company's opening state, then operate it continuously under human supervision — built so the parts that check the work cannot be captured by the parts doing it.

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What it is

Orion has two layers. The **generation layer** turns one sentence into a complete founding business plan — positioning, unit economics, brand, acquisition, operations — using 21 specialised agents across five departments over an enforced append-only state contract. The **operating layer** treats that plan as the company's opening state rather than its output, and runs the same five departments on cadences thereafter.

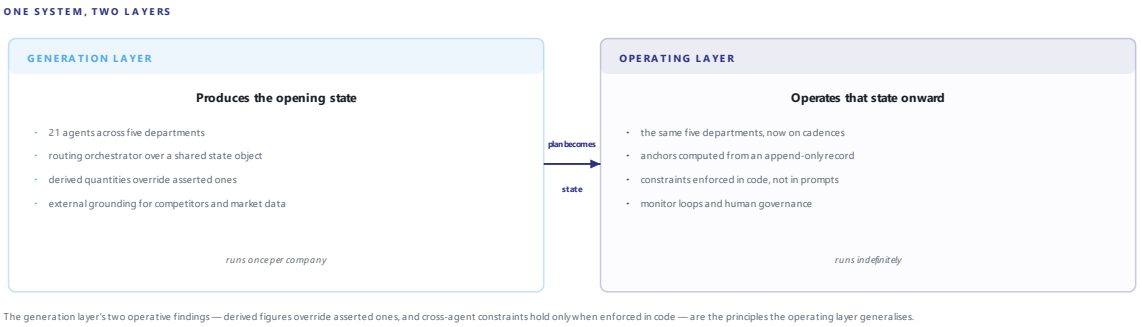


Figure 1 — The generation layer runs once and produces the opening state; the operating layer runs the same five departments indefinitely.

The problem it addresses

Any system that pursues a goal by measuring its progress will eventually improve the measurement rather than the goal. A department rewarded for footfall buys footfall that does not convert; the dashboard stays green while the business deteriorates. Three related failures follow: the objective is never questioned, departments optimise against each other, and thresholds silently expire. In a system operating on daily cadences with nobody in the room, these appear in days rather than quarters.

Two findings from building the generation layer

Prompt-level constraints were insufficient. A market prompt carrying an explicit numeric sanity range, and a revenue prompt stating that year-one revenue must not exceed the obtainable market, were both ignored repeatedly. Five such conflicts were found; all five resolved once the value was computed rather than requested.

Decomposition compounds variance. Over five runs of an identical prompt, the multi-agent pipeline's financial-accuracy standard deviation was 4.9× a single-LLM baseline's, with a 7.6× best-to-worst spread against the baseline's 1.8×. Its best run was the more accurate of the two. The architecture is unreliably better, not reliably worse.

> **BOTH SAY THE SAME THING**
>
> An agent cannot be relied upon to assert a quantity by which it is judged. The generation layer's answer was local — where two agents assert the same value, the derived figure wins. The operating layer generalises it.

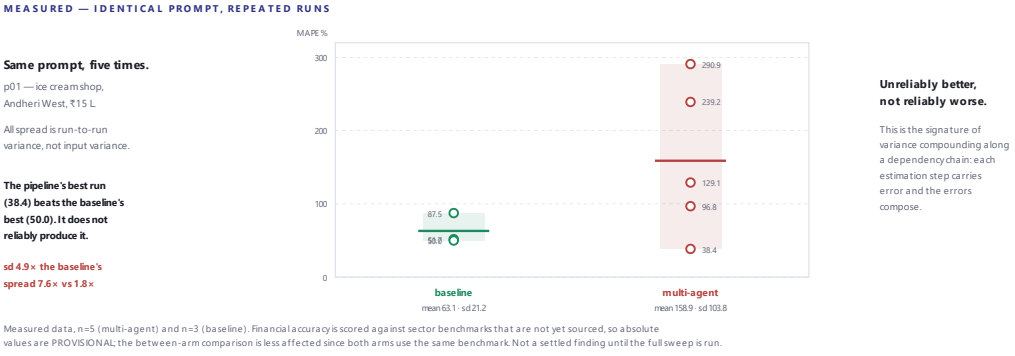


Figure 2 — Five runs of an identical prompt per arm. All spread is run-to-run variance. **Measured**; absolute values provisional pending sourced benchmarks.

How the operating layer answers it

Rather than one loop per department, Orion runs a **graph of loops that check one another**, grounded in three things no department can argue with.

Objective numbers, computed by code. Cash, margin, retention, churn, acquisition cost — derived deterministically from an append-only record of events, never reported by a department.

The Constitution. Hard constraints enforced in the execution path, not described in prompts — spend caps, no data deletion, no irreversible action unapproved. This is the first finding above, applied to authority.

Human judgement. Aesthetics, tone, ethics and strategic direction are routed to a person by classification, not at an agent's discretion. Departments draft; they do not ratify.

THE ANCHOR INVARIANT

No department, planner, worker or reflection process may write an anchor metric, and no learning process may optimise directly against one. Anchors are computed, never reported.

The precedent is the held-out evaluation set in production machine learning: locked, never trained on. If the learner could reach it, the score would stop measuring anything. Same reasoning, applied to a company's measure of itself.

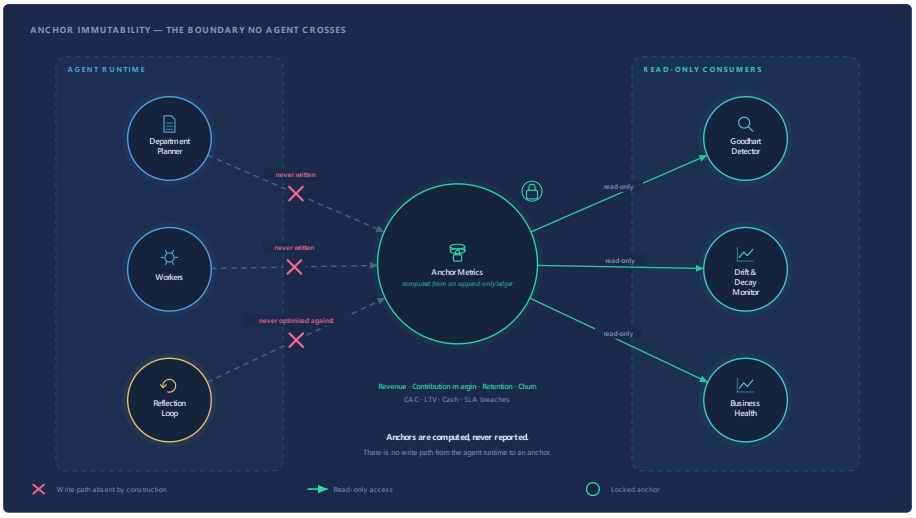


Figure 3 — Anchors are readable by every component and writable by none — not because policy forbids it, but because no such path exists.

Adding oversight is not sufficient on its own. If agent A reviews B, B reviews C and C reviews A, everyone approves everyone and nothing touches reality. Orion therefore validates its own topology at registration: every cycle must contain a code-computed anchor or a human decision, and no monitor may share an evaluation signal with the loop it monitors.

What is built, and what is specified

COMPONENT	STATUS	EVIDENCE
21 agents, five departments	BUILT	Plan completeness 100% of a 26-field rubric vs 80.8% for a single-LLM baseline
Append-only state contract	BUILT	Cross-department writes raise a contract violation; makes agreement measurable at all
Derived-over-asserted reconciliation	BUILT	Resolved all five identified cross-agent conflicts
Dynamic routing	BUILT	17.3% token saving, 0.0 pp completeness delta
External grounding	PARTIAL	OpenStreetMap in use; competitor ground truth not yet collected, so the grounding question is unanswered
Anchors · Constitution · monitor loops · cadences	SPECIFIED	Not yet implemented or measured

Caveats that travel with those numbers. They come from a five-prompt smoke run, not the full thirty-five-prompt sweep. Financial accuracy divides by sector benchmarks that are not yet sourced, so its absolute values are provisional — between-arm comparisons are less affected. Business quality has no oracle comparable to a unit test, so every metric here is a proxy, which is an uncomfortable observation for a system about proxy capture and is stated rather than left implicit.

Where it stands against comparable systems

Orion inherits role specialisation with structured artefacts from MetaGPT, durable execution from workflow engines, approval gates from business process management, and the independent-monitor and locked-evaluation-set patterns from production machine learning. It improves memory from a flat shared context to per-department typed stores, and governance from tool-level permission to business-decision level. What it introduces is the enforced state contract that makes cross-agent disagreement measurable, the anchor invariant, and topology validation that refuses an oversight structure which would only ever agree with itself.

Codebase, database and test artefacts use the identifier `mabgp` *, from the generation layer's original name, retained for continuity with existing test suites. Full architecture document: 16 pp, 15 figures. Research proposal: 14 pp.*